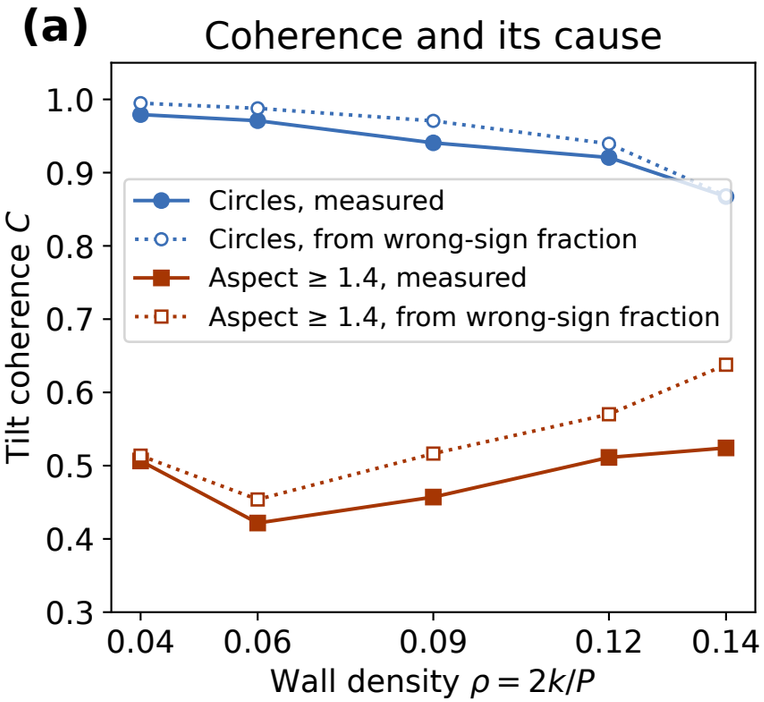
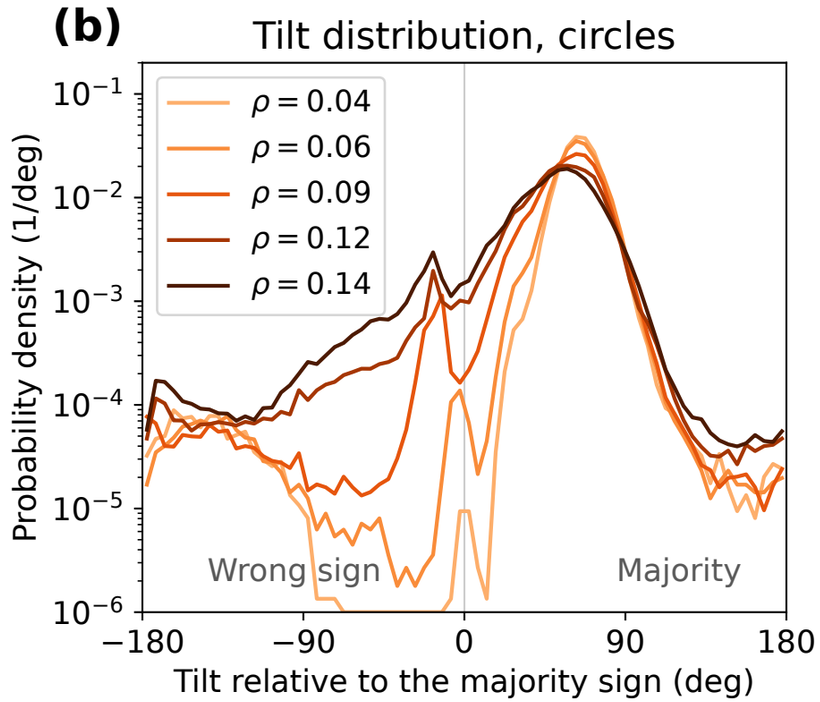
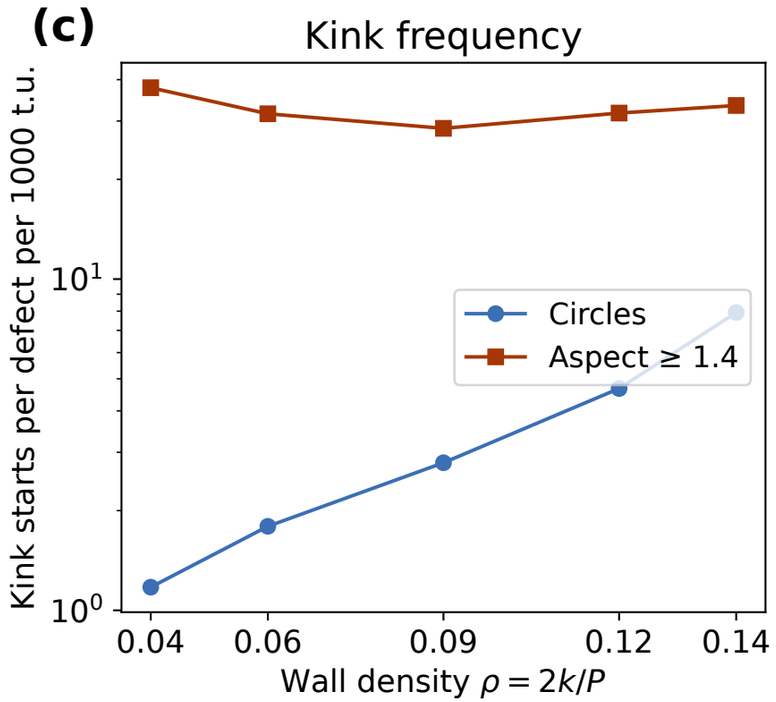
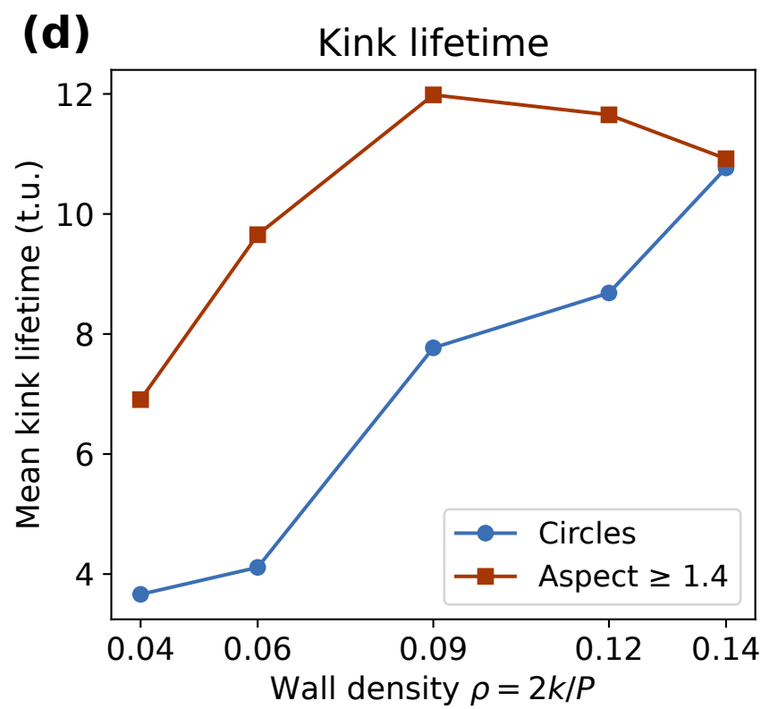
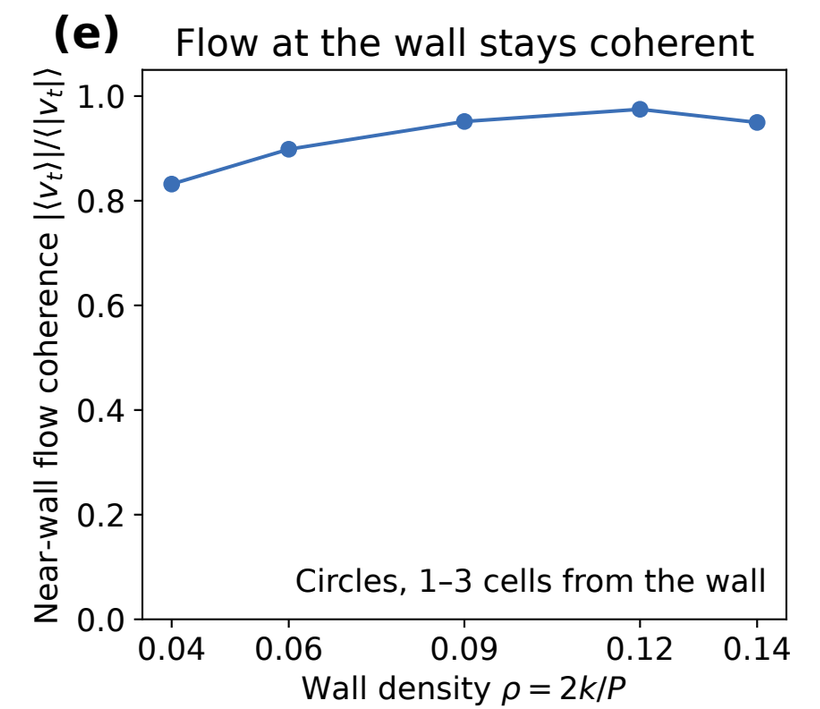
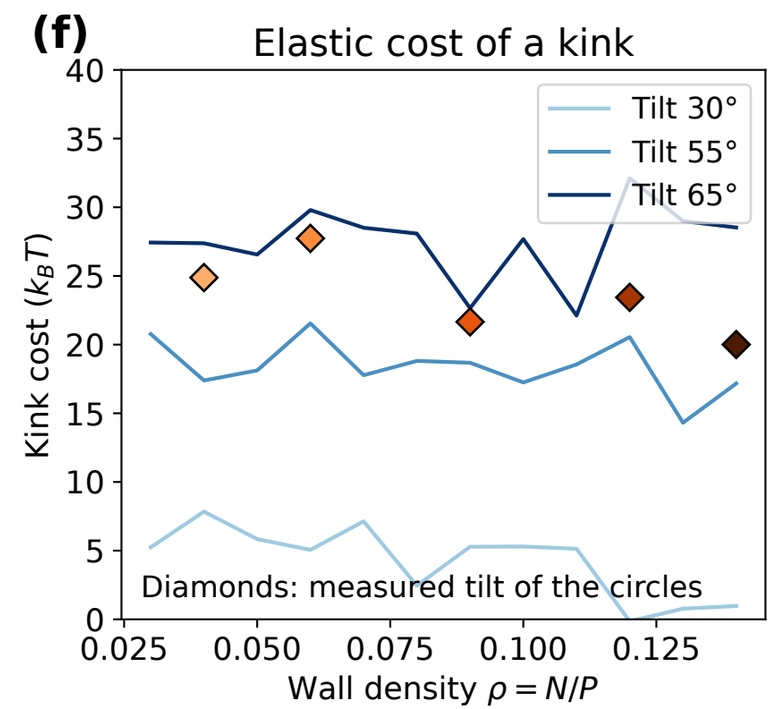

(a)
Coherence and its cause
Tilt coherence C
Circles, measured
Circles, from wrong-sign fraction
Aspect ≥ 1.4, measured
Aspect ≥ 1.4, from wrong-sign fraction
Wall density ρ = 2k/P


(b)
Tilt distribution, circles
Probability density (1/deg)
ρ = 0.04
ρ = 0.06
ρ = 0.09
ρ = 0.12
ρ = 0.14
Wrong sign
Majority
Tilt relative to the majority sign (deg)


(c)
Kink frequency
Kink starts per defect per 1000 t.u.
Circles
Aspect ≥ 1.4
Wall density ρ = 2k/P


(d)
Kink lifetime
Mean kink lifetime (t.u.)
Circles
Aspect ≥ 1.4
Wall density ρ = 2k/P


(e)
Flow at the wall stays coherent
Near-wall flow coherence |⟨v_t⟩|/⟨|v_t|⟩
Circles, 1–3 cells from the wall
Wall density ρ = 2k/P


(f)
Elastic cost of a kink
Kink cost (k_BT)
Tilt 30°
Tilt 55°
Tilt 65°
Diamonds: measured tilt of the circles
Wall density ρ = N/P